

FDK™ Toolkit, Health Domain Fairness Audit

External Validation Report: UCI Diabetes 130-US Hospitals (1999–2008) Dataset

Fourth Real-World Benchmark Validation, Following Justice/COMPAS, Education/Bar Passage, Finance/German Credit & HMDA (Washington 2025)

Dataset: UCI Diabetes 130-US Hospitals (1999–2008)

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Abstract

This report documents the ground-up development and real-dataset validation of the FDK™ (Fairness Diagnostic Kit) Health domain's no-code user experience, following the same methodology established during Finance domain validation. Unlike Finance, Health's no-code dropdown UX did not exist before this validation cycle on either the upload page or the confirmation page; both were built during this work, alongside a pipeline-level binary-validation safeguard and a fallback-detection fix. Validation was conducted against the UCI Diabetes 130-US Hospitals dataset (1999–2008), a freely accessible real-world clinical dataset requiring no credentialed access (unlike MIMIC-IV). Two independent group cuts were audited: race and gender, on the same underlying patient population and toy prediction model. Results mirror the pattern found during HMDA validation: race showed HIGH_BIAS while gender showed LOW_BIAS on the identical dataset and model, reinforcing that a fairness conclusion is specific to the protected attribute tested, not a general property of a dataset or model.

Background

Following the Finance domain's no-code UX redesign and its establishment as the reference implementation for the remaining six FDK domains, Health was audited next. The audit found a materially different starting point than Finance: while Finance already had an unfiltered upload-page dropdown and a static confirmation page, Health's confirmation page (`auto_confirm_health.html`) had no dropdowns whatsoever, a fully static, read-only display stating no manual configuration was required, and the upload page had a hidden, auto-filled target field with no user-facing override at all, despite the backend already being able to accept one. This meant Health required new construction, not just a port of existing fixes, across both the upload and confirmation pages.

Dataset Description

UCI Diabetes 130-US Hospitals for Years 1999-2008

- Source: UCI Machine Learning Repository (freely downloadable, no application or credentialed access required, in contrast to MIMIC-IV, which requires a data-use agreement)
- Represents ten years of clinical care at 130 US hospitals and integrated delivery networks; used in prior fairness literature, including as a Fairlearn tutorial dataset

- Raw extract: 101,766 patient encounters, 50 columns, including race, gender, and age as demographic fields

Preprocessing: the outcome column ("readmitted") is a 3-category string (NO / >30 / <30), not natively binary. Per the standard convention in the fairness literature for this dataset, it was binarized on whether the patient was readmitted within 30 days specifically: `readmitted_30d = 1 if readmitted == '<30' else 0`. Rows with missing-value placeholders in the group columns ("?" for race, "Unknown/Invalid" for gender) were excluded before use as a group cut, leaving 99,492 clean encounters.

Derived prediction column: as with German Credit and HMDA, this dataset contains no native model-decision field. A toy LogisticRegression classifier was trained (features = all other columns excluding identifiers and free-text diagnosis codes, one-hot encoded, including protected attributes as inputs). The first training pass used unweighted classes and produced a functionally degenerate model: 88.8% accuracy but only 1.6% recall on the positive class (predicting positive on just 0.37% of rows against an actual base rate of 11.2%), which would have produced statistically fragile fairness metrics on the smaller subgroups (e.g., 3 predicted positives for the Asian group, $n=641$). Retraining with `class_weight='balanced'` corrected this: 65.7% test accuracy with 53.5% recall on the positive class, and non-degenerate predicted-positive counts across every subgroup.

Findings: No-Code UX Construction & Bug Remediation

UX Construction (new, not a port)

Unlike Finance, no dropdown-based override UX existed for Health before this work. Three functions were built from scratch in `fdk_health.py`, following the same design pattern established for Finance: `_describe_column` (plain-language, role-aware column descriptions, deliberately not applying the "every value unique" identifier-exclusion rule to the probability role, since a genuine probability column naturally has every value unique), `_candidate_columns` (filters each dropdown to statistically plausible columns per role), and `build_column_options` (assembles the four per-role option lists). `auto_confirm_health.html` was rebuilt from a static read-only display into four real, filtered, pre-selected dropdowns, with an explicit warning and a disabled run button when no valid target or prediction column exists, rather than allowing the audit to proceed silently.

Finding 1, Fallback binary-detection gap (found before any real data was tested)

Root cause: the `y_true/y_pred` fallback detection paths used a bare `nunique() == 2` check with no verification that the two values were genuinely binary-coded, while the primary (keyword-matched) detection path already correctly checked for `{0,1}/{1,2}`-style values. Any two-valued non-binary column (e.g. a count field sampled to two distinct values) could reach the fallback path undetected.

Evidence: confirmed with a synthetic column of values `{3,7}` (mimicking a count field), which the unpatched fallback logic accepted as a valid prediction column.

Remediation: both fallback paths now require the same `{0,1}/{0.0,1.0}/{True, False}/{1,2}`-style check already used in the primary detection path. Health's existing convention of treating `{1,2}`-coded columns as valid binary (a common SPSS-style clinical coding convention) was deliberately preserved; this is a domain convention, not a bug, and is more permissive than Finance's equivalent fix, which required strict `{0,1}` only.

Verification: confirmed the same synthetic `{3,7}` column is now correctly rejected, with the pipeline falling through to its documented last-resort behavior (using `y_true` as `y_pred` for baseline pre-implementation tests) instead of silently accepting the corruption-risk column.

Finding 2, Layer-3 pipeline validation gap

Root cause: `fdk_health_pipeline.py`'s own `validate_healthcare_data` function checked required columns, group diversity, and the 20-sample statistical-power floor, but never independently verified that the columns mapped to `y_true/y_pred` were genuinely binary; the equivalent of Finance's third defense layer was missing entirely.

Evidence: confirmed via a direct functional test: a synthetic dataset with a non-binary {3,7}-valued `y_pred` column passed the pre-existing validation checks with `is_valid: true`.

Remediation: added an independent binary-validation check to `validate_healthcare_data`, deliberately redundant with the detection-time checks, so a single bad mapping upstream cannot silently corrupt every downstream metric.

Verification: confirmed the same synthetic {3,7} `y_pred` column is now correctly rejected with `is_valid: false` and a descriptive error, while genuine {0,1} data continues to pass.

Finding 3, Confirmation-page override inputs were already present but silently dead

Root cause: `auto_confirm_health.html` already submitted hidden form fields (`group_col`, `y_true_col`, `y_pred_col`, `y_prob_col`) to the `run-audit` route, but `run_health_audit_with_mapping` never read `request.args` at all; it only used whatever mapping was stored in session from the original auto-detection. Any override submitted through the form was silently discarded.

Remediation: the four hidden inputs were converted into real, visible, filtered dropdowns (see UX Construction above), and `run_health_audit_with_mapping` was updated to read and apply each override from `request.args`, re-validating each one against its role's own valid-candidate list before acceptance (so a tampered or stale request cannot re-introduce an invalid column).

Verification: confirmed via the two live audit runs in this report that the group column (race, then gender) submitted through the new dropdown was correctly applied and reflected in each resulting report's `column_mapping` metadata.

Finding 4, Upload page had no manual target override at all

Root cause: unlike Finance's upload page (which has an unfiltered target dropdown, a deliberate but separate limitation), Health's upload page had only a hidden input auto-filled from the shared `/api/detect-columns` endpoint's single `suggested_target` value, with no user-facing control to change it before submission.

Remediation: the hidden input was converted into a visible dropdown, populated by parsing the chosen file's header row directly in the browser (independent of the shared cross-domain endpoint's response shape, which was not visible for direct modification), pre-selected to the server's suggested target once that response arrives.

Verification: confirmed working directly by the user in their deployed environment.

Finding 5, Regression introduced and self-corrected during this validation

What happened: an edit made during Finding 1's remediation accidentally deleted the function signature and header comment for `build_human_summaries` (an existing, unrelated summary-generation function), leaving its body orphaned with no `def` statement. This was not caught by syntax checking alone, since the file remained syntactically valid Python; it surfaced only as a runtime `NameError` once a full audit was run end-to-end ("Health Audit Failed: name 'build_human_summaries' is not defined").

Detection: the user encountered the error directly while testing the deployed confirmation-page flow.

Remediation: the missing function signature and header comment were restored exactly as they existed in the original file, verified by re-listing every top-level function/class definition in the file against the original 17 and confirming all were present plus the 3 legitimate additions, and by directly calling `build_human_summaries` to confirm it executes without error.

Lesson for future domain rollouts: when a `str_replace`-style edit uses a following function's signature line purely as an anchor for uniqueness, that signature line must be explicitly reproduced in the replacement text, not assumed to survive untouched; anchor text is part of what gets replaced, not a boundary that is preserved automatically.

Fairness Findings

Diabetes 130-US Hospitals, race

Group	n	Selection rate	Base rate
AfricanAmerican	19,210	0.3667	0.1122
Caucasian	76,099	0.3513	0.1129
Hispanic	2,037	0.2695	0.1041
Asian	641	0.2590	0.1014
Other	1,505	0.1907	0.0963

- Statistical parity difference: 0.1760 (AfricanAmerican selection rate 0.3667 vs. Other 0.1907)
- Demographic parity ratio: 0.5200
- Equal opportunity (TPR) difference: 0.2056; FNR difference: 0.2056, flagged by the pipeline itself as a critical, patient-safety-relevant finding (a missed true positive here means a real readmission risk goes uncaught)
- Composite bias score: 0.1596, overall assessment: HIGH_BIAS (“urgent review needed”)

Notable pattern: base (actual readmission) rates are close together across all five groups (9.6%–11.3%), while predicted selection rates diverge sharply (19.1%–36.7%); the model’s predictions are far less consistent across race groups than the underlying reality they are meant to reflect.

Diabetes 130-US Hospitals, gender

Group	n	Selection rate	Base rate
Female	53,575	0.3501	0.1131
Male	45,917	0.3489	0.1112

- Statistical parity difference: 0.0012
- Demographic parity ratio: 0.9966, comfortably close to parity
- Composite bias score: 0.0043, overall assessment: LOW_BIAS (“no critical fairness issues detected”)

Cross-dataset observation

The same Diabetes 130-US Hospitals population and toy model that showed HIGH_BIAS on the race cut (composite bias score 0.1596, demographic parity ratio 0.520) showed LOW_BIAS on the gender cut (composite bias score 0.0043, demographic parity ratio 0.997). This directly replicates the pattern found during HMDA validation in the Finance domain (race: HIGH_BIAS, sex: LOW_BIAS, same population), now confirmed on a second, independent dataset in a different domain. Auditing only one protected attribute on a given dataset does not license a general conclusion about that dataset or model’s fairness; the finding is specific to the attribute tested.

Scope Limitations

- The dataset contains no native model-decision column; a toy LogisticRegression classifier was trained to derive predicted outcomes. These fairness findings describe the toy model’s behavior, not any real deployed hospital readmission-risk model; they validate that FDK’s pipeline correctly detects and quantifies disparities, not that any specific real-world model or hospital system exhibits these disparities.
- The toy classifier required a class-weighting correction after an initial degenerate fit (1.6% recall on the positive class); the corrected model has non-trivial disagreement (53.5% recall) but still trades away raw

accuracy (65.7%) for balanced learning across classes, which should be read as a deliberate methodological choice for fairness-testing purposes, not an optimized production model.

- The Health confirmation-page dropdown UX and upload-page manual override did not exist before this validation cycle and are new construction, not a port of previously shipped Finance code; they have been verified against one real dataset so far, not the breadth of edge cases Finance has now seen.
- The "Other" race category (n=1,505) and "Asian" (n=641) are the thinnest subgroups in the race cut; both clear the project's 20-sample statistical-power floor by a wide margin, but findings on these groups specifically carry more estimation variance than the larger Caucasian/AfricanAmerican groups.
- This dataset spans 1999–2008 clinical practice at 130 US hospitals; findings should not be generalized to current clinical practice or to healthcare systems outside the United States without further testing.

Conclusion

The FDK Health domain's no-code UX, built from scratch during this validation cycle rather than ported from an existing implementation, along with its fallback-detection fix and new pipeline-level binary validation, are verified working end-to-end against a real-world clinical dataset. As with Finance, the validation confirms both that the pipeline correctly computes and reports fairness metrics against genuine data, and that a single dataset can yield materially different fairness conclusions depending on which protected attribute is audited. This is now the second domain, after Finance, to independently reproduce that same pattern (HMDA race/sex; Diabetes race/gender), strengthening the case that FDK's multi-group-cut capability addresses a real and recurring analytical need rather than a one-off finding. Health is now validated as a working reference implementation alongside Finance for the remaining five FDK domains.